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A few logs suffice to build (almost) all trees (I)

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Abstract

A phylogenetic tree, also called an “evolutionary tree,” is a leaf-labeled tree which represents the evolutionary history for a set of species, and the construction of such trees is a fundamental problem in biology. Here we address the issue of how many sequence sites are required in order to recover the tree with high probability when the sites evolve under standard Markov-style i.i.d. mutation models. We provide analytic upper and lower bounds for the required sequence length, by developing a new polynomial time algorithm. In particular, we show when the mutation probabilities are bounded the required sequence length can grow surprisingly slowly (a power of log n) in the number n of sequences, for almost all trees.

Real-World Impact & Application

Reconstructing Ancient Dinosaurs from Tiny DNA Fragments: When evolutionary biologists only have short, degraded DNA fragments, how can they accurately place an extinct species on the tree of life? This paper proved mathematically that even surprisingly short genetic sequences are enough to build accurate evolutionary trees.